

Hsuan-Fu (Jeff) Hsu

Software Engineer | Backend, Full-Stack & Cloud Systems
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EDUCATION

University of Southern California

Master of Science in Computer Science

- Relevant Coursework: Analysis of Algorithms, Distributed Database Systems, Web Technologies

National Taipei University (NTPU)

Bachelor of Science in Computer Science

Los Angeles, CA

Jan 2026 - May 2027 (Expected)

New Taipei City, Taiwan

Sept 2020 - June 2024

EXPERIENCE

Tech-Management Workshop Co., Ltd.

Software Engineer Intern

Taichung, Taiwan

July 2026 - Aug 2026

- Independently delivered a **government birth-grant module** from procurement specifications without access to legacy source code using **C#, .NET 8, ASP.NET Core MVC, Entity Framework Core, and SQL Server**.
- Implemented **three end-to-end workflows** - Taiwan national ID and eligibility validation, dynamic QR code generation, and two-page A4 printing - for a system designed to serve **29 municipal offices**.
- Added **three deployment-readiness controls** - server-side validation, anti-forgery protection, and HTTPS/HSTS - then packaged the module for the company intranet and resolved defects found through cross-browser and functional testing.

National Taipei University

Undergraduate Researcher - Advisor: Prof. Bor-Shing Lin

New Taipei City, Taiwan

July 2022 - Feb 2024

- Owned the **Python** signal-processing and data-preparation pipeline that converted raw photoplethysmography (**PPG**) signals into model-ready datasets for a **three-person research team**.
- Implemented **0.5-10 Hz Bessel filtering**, signal-quality screening, feature selection, and **LOSO cross-validation**; the dual-band regression model achieved **10.3% MAPE**.
- Received a national undergraduate research fellowship and earned fourth place in NTPU project competition.

TECHNICAL SKILLS

Languages: C#, C++, Python, TypeScript, JavaScript, SQL

Backend & Web: ASP.NET Core Web API/MVC, Entity Framework Core, Vue 3, Node.js, Express, REST APIs

Cloud, Data & DevOps: Docker, Google Cloud Run, SQL Server, SQLite, GitHub Actions, CI/CD

Testing & Systems: xUnit, integration testing, Git, Linux, OpenGL, GLSL, NumPy

PROJECTS

Task Management System | ASP.NET Core, Vue 3, TypeScript, EF Core, Docker | [GitHub](#)

Jan 2026 - Sept 2026

- Built a multi-user task platform with **RESTful APIs**, server-side search, filtering, sorting, pagination, dashboard aggregates, and a typed **Vue 3/TypeScript** interface.
- Secured user data with **password hashing**, **JWT authentication** using **HttpOnly cookies** with **SameSite protection**, per-user ownership filters, rate limiting, and structured Problem Details responses.
- Added **xUnit integration tests**, **GitHub Actions CI**, database health checks, and a **multi-stage Docker build** that compiles and serves the frontend and API from one origin.

Art Explorer | Node.js, Express, Docker, Google Cloud Run | [Live](#) | [GitHub](#)

Jan 2026 - June 2026

- Built and deployed an art discovery service integrating the **Met Museum**, **Harvard Art Museums**, and **Wikipedia APIs** with pagination, interactive maps, related works, and persistent favorites.
- Designed an **Express** aggregation and proxy layer, containerized the application with **Docker**, and deployed it to **Google Cloud Run**.

3D Rendering Engine | C++, OpenGL, GLSL | [GitHub](#)

Sept 2023 - Dec 2023

- Built a custom **OBJ/MTL parser** and real-time **OpenGL** pipeline using **VAOs/VBOs**, MVP transformations, texture mapping, and multi-light Phong shading.
- Implemented material loading and interactive scene rendering with **GLSL shaders** and **Blender**-prepared models and assets.

Online Judge Platform | Vue.js, REST APIs | [GitHub](#)

Sept 2021 - Jan 2022

- Led a **four-developer frontend team**, defined **REST API contracts** with backend and dispatcher teams, and built asynchronous submission tracking for a distributed code-evaluation platform.

PUBLICATION

Hsu Hsuan-Fu et al., "Noninvasive Portable System With Dual Near-Infrared Light Technology for Monitoring Uric Acid Concentration," IEEE Transactions on Human-Machine Systems, accepted for publication, 2026.